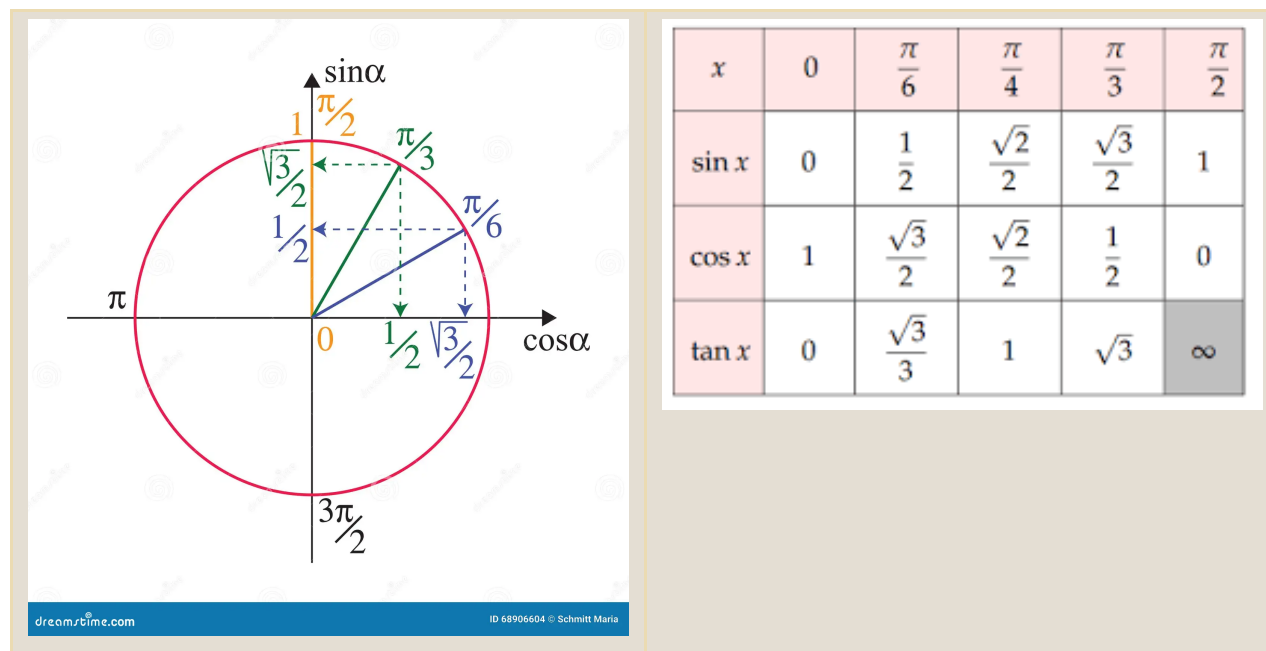


# Formules trigo



- $\sin^2(\theta) + \cos^2(\theta) = 1$

## Symétrie

|                                                 |                            |
|-------------------------------------------------|----------------------------|
| $\cos\left(x + \frac{\pi}{2}\right) = -\sin(x)$ | $\cos(x + \pi) = -\cos(x)$ |
| $\cos\left(\frac{\pi}{2} - x\right) = \sin(x)$  | $\cos(\pi - x) = -\cos(x)$ |
| $\sin\left(x + \frac{\pi}{2}\right) = \cos(x)$  | $\sin(x + \pi) = -\sin(x)$ |
| $\sin\left(\frac{\pi}{2} - x\right) = \cos(x)$  | $\sin(\pi - x) = \sin(x)$  |

|                      |                       |
|----------------------|-----------------------|
| $\cos(-x) = \cos(x)$ | $\sin(-x) = -\sin(x)$ |
|----------------------|-----------------------|

## Sommes

- $\cos(a + b) = \cos(a)\cos(b) - \sin(a)\sin(b)$
- $\cos(a - b) = \cos(a)\cos(b) + \sin(a)\sin(b)$
- $\sin(a + b) = \sin(a)\cos(b) + \cos(a)\sin(b)$
- $\sin(a - b) = \sin(a)\cos(b) - \cos(a)\sin(b)$
- $\tan(a + b) = \frac{\tan(a) + \tan(b)}{1 - \tan(a)\tan(b)}$
- $\tan(a - b) = \frac{\tan(a) - \tan(b)}{1 + \tan(a)\tan(b)}$

## Angles doubles et demi-angle

- $\cos(2x) = 2 \cos^2(x) - 1 = 1 - 2 \sin^2(x) = \cos^2(x) - \sin^2(x)$
- $\sin(2x) = 2 \cos(x) \sin(x)$
- $\tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$

## Produits $\rightarrow$ sommes

- $\cos(a) \cos(b) = \frac{1}{2}(\cos(a+b) + \cos(a-b))$
- $\sin(a) \sin(b) = \frac{1}{2}(\cos(a-b) - \cos(a+b))$
- $\sin(a) \cos(b) = \frac{1}{2}(\sin(a+b) + \sin(a-b))$

## Autres

- $\tan'(x) = 1 + \tan^2(x) = \frac{1}{\cos^2(x)}$
- $\tan'(u(x)) = u'(x)(1 + \tan^2(u(x))) = \frac{u'(x)}{\cos^2(u(x))}$
- $a \cos(t) + b \sin(t) = A \cos(t + \varphi)$  avec :
  - $z = a + ib$
  - $A = |z| = \sqrt{a^2 + b^2}$
  - $\varphi = \arg(z) = \arctan\left(\frac{b}{a}\right)$